# Lab 05 — Data Distribution Analysis

**1. Lab Title**

**Understanding Data Distribution for Machine Learning**

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**2. Lab Objective**

After completing this lab students will be able to:

1. Understand **statistical distributions used in machine learning**.
2. Calculate **mean, median, and variance** of a dataset.
3. Visualize data distribution using graphs.
4. Interpret whether data follows a **normal distribution**.

These statistics are foundational in machine learning because algorithms often assume specific distributions of data (for example, Gaussian assumptions in many probabilistic models).

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**3. Background Theory**

**3.1 What is Data Distribution?**

A **data distribution** shows how values are spread across a dataset.

Example:

Student Marks
45, 50, 55, 60, 70, 80, 90

The distribution tells us:

- Are most values **low or high**
- Are values **clustered together**
- Are there **outliers**

Understanding distribution helps ML models understand **patterns in data**.

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### 3.4 Variance

Variance measures **how spread out the data is**.

Formula

[
Variance = \frac{\sum (x - \mu)^2}{n}
]

Where

- (x) = each value
- (\mu) = mean

If variance is:

Small → data values are **close to mean**
Large → data values are **spread out**

Source: *An Introduction to Statistical Learning*.

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### 3.5 Normal Distribution (Gaussian Distribution)

A **normal distribution** is a symmetric bell-shaped curve.

Properties:

- Mean = Median = Mode
- Most values are near the center
- Few values are very high or very low

Many ML algorithms assume normally distributed data.

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### 4. Software / Tools Required

Students must install:

- Python

- Jupyter Notebook or VS Code
- Libraries:

numpy

pandas

matplotlib

Install libraries:

pip install numpy pandas matplotlib

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**5. Dataset for Lab**

Students will analyze **student marks dataset**.

Create a CSV file called:

student_marks.csv

Example data:

| Student | Marks |
|---|---|
| A | 45 |
| B | 60 |
| C | 55 |
| D | 70 |
| E | 65 |
| F | 80 |
| G | 75 |
| H | 50 |
| I | 90 |
| J | 85 |

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**6. Lab Procedure (Step-by-Step)**

**Step 1 — Import Libraries**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Explanation:

- **pandas** → used for dataset handling
- **numpy** → used for statistical calculations
- **matplotlib** → used for visualization

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**Step 2 — Load Dataset**

```
data = pd.read_csv("student_marks.csv")
print(data)
```

Output example

| Student | Marks |
|---|---|
| A | 45 |
| B | 60 |

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**Step 3 — Calculate Mean**

```
mean_marks = data["Marks"].mean()
print("Mean:", mean_marks)
```

This calculates average marks.

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**Step 4 — Calculate Median**

```
median_marks = data["Marks"].median()
```

```
print("Median:", median_marks)
```

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**Step 5 — Calculate Variance**

```
variance_marks = data["Marks"].var()
```

```
print("Variance:", variance_marks)
```

Variance calculation in pandas follows standard statistical formula.

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**Step 6 — Visualize Data Distribution**

**Histogram**

```
plt.hist(data["Marks"], bins=5)
```

```
plt.title("Marks Distribution")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

Histogram shows how marks are distributed.

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**Step 7 — Box Plot**

```
plt.boxplot(data["Marks"])
```

```
plt.title("Box Plot of Marks")
```

```
plt.show()
```

Box plot helps identify:

- median
- outliers
- data spread.

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**7. Lab Task (Student Work)**

Each student must create **their own dataset** :

Students must:

1. Create dataset with **10 students including themselves**.
2. Calculate

- Mean
- Median
- Variance

3. Plot

- Histogram
- Boxplot

4. Write **2–3 lines explaining the distribution**.

Example explanation:

"Most marks are between 60–80 which shows the distribution is moderately centered."

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**8. Expected Output**

Students should submit:

1. Python Notebook / Python file
2. Dataset CSV
3. Screenshots of graphs
4. Short interpretation of distribution

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